

Challenge: Improving emergency response and resource allocation after major earthquakes

1. News article



This news article, “Japan’s Noto earthquake: thousands of survivors struggle as accusations of neglect grow,” was written by Justin McCurry and published on 14 February 2024 in The Guardian, a British daily newspaper and internationally recognized news organization.

2. Main challenge

The main challenge described in the chosen article is the difficulty of providing fast and effective support to communities after such major events like the earthquake from 1st January 2024 in Japan. The article shows that the earthquake caused significant difficulties with evacuation conditions, access to water, damaged infrastructure and most importantly the loss of human lives.

3. Relevance

This topic was chosen, because earthquakes still represent one of the major natural hazards that affects societies all around the world. The most affected countries are Japan, Indonesia, Chile, China, Iran etc. They continue to experience destructive earthquakes that often cause deaths, damage to infrastructure, displacement, and long term social and economic consequences. We decided to focus on Japan, since it shows that even countries with strong preparedness systems can still face serious consequences and have difficulties supporting the affected communities after major earthquakes. In addition, Japan faces the ongoing risk of a major Nankai Trough earthquake, which could have a potentially devastating impact on many communities in the west part of Japan. This makes improving

emergency response and the use of timely and accurate information particularly important for future disaster preparedness.

4. Stakeholders

The primary stakeholders of the challenge are victims and their relatives, who need timely information and access to essential resources. Emergency aid organizations can use satellite data to quickly identify the most severely affected areas and improve the distribution of food and non-food items. Authorities can use the collected data to make more informed decisions about recovery and rebuilding as they prepare for the possibility of future events. Researchers and scientific organizations are also stakeholders, as satellite observations provide valuable information about changes to the affected area.

5. Scientific peer-reviewed articles

Scientific article 1 (Original research article)

Li, Q., Zhang, J., & Jiang, H. (2024). Incorporating multi-source remote sensing in the detection of earthquake-damaged buildings based on logistic regression modelling. *Heliyon*. <https://doi.org/10.1016/j.heliyon.2024.e32851>

This paper proposes an understanding of how satellite data can help evaluate earthquake damage to buildings based on multi-source remote sensing data. This model achieves the evaluation of buildings with different degrees of damage by defining the quantitative relationship between the earthquake damage assessment index and image features. The work provides several steps for addressing the problem, including image processing and feature performance evaluation.

By comparing with the results of on-site investigations, the extraction accuracy is 94.2 %, and the area under the ROC curve of the model is 0.827. Model and accuracy verification show that the model has high reliability and accuracy and can be used for the earthquake damaged building assessment. If there are new data sources in the future, this method can also be used to construct models and evaluate seismic damage to buildings.

Scientific article 2 (Scientific Report)

Yuki Orihashi, Shingo Suzuki, Hiroyuki Kuramoto, Yoshihiro Ura & Go Urakawa(2024). The Actual State of Data Utilization in Disaster Response in Municipalities: Collaborative Practice in Wajima City, Following the 2024 Noto Peninsula Earthquake. *Journal of Japan Society of Natural Disaster Science*, 43(3), 619–630. https://doi.org/10.24762/jnds.43.3_619

This study examined how GIS and data were used to support disaster response in Wajima City after the 2024 Noto Peninsula Earthquake. The researchers worked with the

municipality and external support organizations to collect, process, visualize, and share disaster-related information using a common GIS platform.

The study showed that GIS helped the city collect information efficiently and visualize important data for disaster response. For example, information on building damage, evacuation shelters, temporary housing, and water supply recovery was collected and displayed using maps and dashboards. This information was used to support information sharing, decision-making, and progress management. The study also showed that when some necessary information was not yet available, other available data could sometimes be used as a substitute to support rapid decision-making.

This is relevant to our societal challenge because timely and accurate information is essential for effective emergency response and resource allocation after earthquakes. The study provides a real-world example from the 2024 Noto Peninsula Earthquake of how GIS can help municipalities understand the situation, identify needs, and make decisions more efficiently.

Scientific article 3 (Original research article)

[Timothy Tiggeloven , Davide Mauro Ferrario , Judith N. Claassen , Wiebke S. Jäger , Yuliya Shapovalova , Maki Koyama , Marleen C. de Ruiter , James E. Daniell , Silvia Torresan , and Philip J. Ward.] **A Global Approach for Mapping Multi-Hazard Susceptibility Using Deep Learning: A Case Study in Japan.** (2025)

Journal: *Artificial Intelligence for Earth Systems*

<https://doi.org/10.1175/AIES-D-25-0039.1>

I found this article relevant to our societal challenge because it shows how data science and deep learning are used to identify vulnerable areas in Japan to natural hazards, such as earthquakes. In this study, deep learning and geospatial data were used to create susceptibility maps showing areas with higher levels of risk. The model combines spatial hazard data for nine hazards across Japan. For earthquakes the model uses Japan's seismic hazard map based on recorder ground acceleration and 30 year exceedance probability. So based on all of this information they can better plan emergency resources, protect infrastructure and come up with evacuation strategies.

Scientific article 4 (Original research article/Case study)

Ishiwatari, M. (2024). Leveraging Drones for Effective Disaster Management: A Comprehensive Analysis of the 2024 Noto Peninsula Earthquake Case in Japan. *Progress in Disaster Science*, 23, 100348.

<https://doi.org/10.1016/j.pdisas.2024.100348>

This article evaluates drone usage in the 2024 Noto Peninsula Earthquake Case in Japan. It discusses the benefits of drones and the improvements that can still be made. The drones were used for:

1. Delivering food, medicine and oil
2. Restore communication, where power systems are disrupted. This way there can be communication through cellphones about where help is needed.
3. Finding damaged and dangerous locations, to help the rescue teams

The discussion points are:

1. The lack of integration into disaster plans, everything was done on the spot instead of it being thought out beforehand. This also counts for being allowed to use drones.
2. The collected data should be integrated faster into the Disaster Information Systems.
3. The government should make clear arrangements with companies on drone usage after earthquakes.

So this article concludes that the usage of drones after earthquakes has a lot of advantages and improvement in emergency response and they should be used.